THE SUM OF THE ORDERS OF THE TAME SYMBOLS OVER THE HEIGHT ONE PRIMES OF A TWO DIMENSIONAL NOETHERIAN LOCAL DOMAIN VANISHES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement opens with a reference block recording that for an excellent ring the result is Kato's Milnor K-theory proposition; the block is retained verbatim inside the counted span, and the chapter preamble carries the reference environment as a comment environment so it does not print.

Editorial note. Canonical location of the counted proposition: Stacks tag 02QJ (sources/stacks-chow/pinned-original-source.tex line 18340).

Allowed standard prerequisites (editorial).

- Additivity of the tame symbol and additivity of the order functions. Same chapter, Lemma labelled lemma-multiplicativity-symbol (source line 17434) and Algebra, Lemma labelled algebra-lemma-ord-additive (Stacks tag 02MC), invoked at proof lines 18370-18373. Supply the reduction of the general case f, g in K^* to the case f = a, g = b in A, which is the first step of the proof.
- The determinant of a two-periodic complex and its equality with the order sum. Same chapter, Proposition labelled proposition-length-determinant-periodic-complex (source line 18024), invoked at proof line 18381. Supplies the equivalence of the displayed order sum with the vanishing of e_A(A/(ab), a, b); it is retained only as a tagged navigation pointer because its interval lies inside a public item.
- Definition of the order function on the residue field of a one dimensional quotient. Algebra, Definition labelled algebra-definition-ord (Stacks tag 02MD), used implicitly through the order functions appearing in the statement. Fixes the meaning of ord_{A/q} in both displayed sums.

Source reference. Original lemma 02Q2: sources/stacks-chow/pinned-original-source.tex, source lines 17483–17483. Same-chapter pointer into the author environment lemma at source line 17483; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02Q0: sources/stacks-chow/pinned-original-source.tex, source lines 17434–17434. Same-chapter pointer into the author environment lemma

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at source line 17434; invoked at proof line 18371 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02MC: sources/stacks-chow/tags.txt, source lines 3439–3439. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 18373 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original proposition 02QD: sources/stacks-chow/pinned-original-source.tex, source lines 18024–18024. Same-chapter pointer into the author environment proposition at source line 18024; invoked at proof line 18381 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1 (Key Lemma)**.** *Let A be a 2-dimensional Noetherian local domain with fraction field K. Let $f, g \in K^*$. Let $\mathfrak{q}_1, \ldots, \mathfrak{q}_t$ be the height 1 primes $\mathfrak{q}$ of A such that either f or g is not an element of $A_{\mathfrak{q}}^*$. Then we have*

$$\sum\nolimits_{i=1,\ldots,t} ord_{A/\mathfrak{q}_i}(d_{A_{\mathfrak{q}_i}}(f, g)) = 0$$

We can also write this as

$$\sum\nolimits_{height(\mathfrak{q})=1} ord_{A/\mathfrak{q}}(d_{A_{\mathfrak{q}}}(f, g)) = 0$$

since at any height one prime $\mathfrak{q}$ of A where $f, g \in A_{\mathfrak{q}}^*$ we have $d_{A_{\mathfrak{q}}}(f, g) = 1$ by Lemma 02Q2.

Proof. Since the tame symbols $d_{A_{\mathfrak{q}}}(f, g)$ are additive (Lemma 02Q0) and the order functions $\mathrm{ord}_{A/\mathfrak{q}}$ are additive (Algebra, Lemma 02MC) it suffices to prove the formula when $f = a \in A$ and $g = b \in A$. In this case we see that we have to show

$$\sum\nolimits_{\mathrm{height}(\mathfrak{q})=1} \mathrm{ord}_{A/\mathfrak{q}}(\det_\kappa(A_{\mathfrak{q}}/(ab), a, b)) = 0$$

By Proposition 02QD this is equivalent to showing that

$$e_A(A/(ab), a, b) = 0.$$

Since the complex $A/(ab) \xrightarrow{a} A/(ab) \xrightarrow{b} A/(ab) \xrightarrow{a} A/(ab)$ is exact we win. □